Chapter 10

Troubleshooting

Fault sort explanation

MONT71 has almost 60 pieces of protection functions.

MONT71 monitors all kinds of input signal, running condition etc. If some abnormal error happens, relevant fault protection functions will act and the controller will display the fault code.

Error information produced by MONT71 can be divided into 3 sorts according to their influence to the system. Different fault has different disposal mode, which is as shown in the next table.

Fault sort	Relevant disposal	Remark
Level 1 fault	• Display fault code • Error relay output action	Any kind of working condition will not be influenced
Level 2 fault	• Display fault code • Error relay output action • Stop at the nearest landing when in distance control, then stop running • Stop running at once in other work condition	After stop, the system will close off output at once, and close brake
Level 3 fault	• Display fault code • Error relay output action • The system blank off output at once, close brake and stop running	Forbid running

Fault reset method

After the fault is removed, you can do fault reset through the following ways:

1. Reset through the panel.
2. Make MONT71 completely power-down.
3. Some faults may auto-reset.

Fault code description

The fault's display code, cause, countermeasure and sort are seen in Table 10-1.

The panel displays five data: E+ Fault code

The panel can prompt fault code causes and countermeasures, and the detail operation is referred to section 7.1.3 (page 98).

The MCB's small panel displays three data: E+ Fault code

Table 10-1 Fault content and countermeasures

<table>
<tr><th>Panel display</th><th>Small panel display</th><th>Fault name</th><th>Fault cause</th><th>Countermeasure</th><th>Sort</th></tr>
<tr><td>Lu</td><td>Lu</td><td>DC bus undervoltage</td><td>1: Power-on initial state, power-down end state
2: Input voltage is too low
3: Wiring does not regulate resulting in hardware undervoltage
4: Model is set incorrectly</td><td>1: Normal power up/down status, normal correctly
2: Check the input supply voltage
3: Check the wiring and regulate it
4: Set the model (Y00.01) correctly</td><td>3</td></tr>
<tr><td>E0001</td><td>E01</td><td>Controller output Acc overcurrent</td><td>1: Main circuit output is grounding
2: Main circuit output is short wiring
3: The motor has not done parameter auto-tuning
4: Load is too heavy
5: Encoder signal is wrong
6: Encoder signal interference is serious
7: Acceleration curve is too steep</td><td rowspan="3">1: Check the main circuit output side whether groud is short-circuited and output phase is short-circuited
2: Check whether the power wiring is damaged and the wiring is solid
3: Check wheter the motor internal exists a short circuit or shorted to ground
4: Outputside contactor is abnormal
5: Star-delta contactor causes MONT71 output short-circuited
6: Set the correct motor parameters (Group F07 / Group F10)
7: Restart motor parameter auto-tuning (Group F07 / Group F10)
8: Check whether the brake is abnormal
9: Check whether the mechanical is stuck
10: Check whether the elevator balance coefficient is correct
11: Check whether the encoder wiring is reliable
12: Set the correct encoder parameters (Group F11)
13: Encoder is installed reliably
14: Check whether the encoder alignment is independence wear tube, trace distance is too long and the shielded cable is single-end grounded
15: Check whether the acceleration /deceleration curve (Group F03) is too large</td><td>3</td></tr>
<tr><td>E0002</td><td>E02</td><td>Controller output Dec overcurrent</td><td>1: Main circuit output is grounding
2: Main circuit output is short wiring
3: The motor has not done parameter auto-tuning
4: Load is too heavy
5: Encoder signal is wrong
6: Encoder signal interference is serious
7: Deceleration curve is too steep</td><td>3</td></tr>
<tr><td>E0003</td><td>E03</td><td>Controller output constant speed overcurrent</td><td>1: Main circuit output is grounding
2: Main circuit output is short wiring
3: The motor has not done parameter auto-tuning
4: Load is too heavy
5: Encoder signal is wrong
6: Encoder signal interference is serious</td><td>3</td></tr>
</table>

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<tr><th>Panel display</th><th>Small panel display</th><th>Fault name</th><th>Fault cause</th><th>Countermeasure</th><th>Sort</th></tr>
<tr><td>E0004</td><td>E04</td><td>DC bus voltage Acc overvoltage</td><td>1: Input voltage is too high
2: Acceleration curve is too steep
3: Brake resistance is too much
4: Braking unit is abnormal
5: Power feedback is abnormal</td><td rowspan="3">1: Adjust the input voltage, check whether the bus voltage (D01.06) is normal
2: Check the balance coefficient
3: Select the appropriate braking resistor, refer to section 5.6 Braking Resistor Selection (page 53)
4: Connect with braking unit or power regenerative unit, check the related equipment</td><td>3</td></tr>
<tr><td>E0005</td><td>E05</td><td>DC bus voltage Dec overvoltage</td><td>1: Input voltage is too high
2: Deceleration curve is too steep
3: Brake resistance is too much
4: Braking unit is abnormal
5: Power feedback is abnormal</td><td>3</td></tr>
<tr><td>E0006</td><td>E06</td><td>DC bus voltage constant speed overvoltage</td><td>1: Input voltage is too high
2: Brake resistance is too much
3: Braking unit is abnormal
4: Power feedback is abnormal</td><td>3</td></tr>
<tr><td>E0008</td><td>E08</td><td>Power module faulty</td><td>1: Short circuit between phases output or the ground
2: Motor wiring is too long
3: Work environment is overheating
4: Power module is damaged</td><td>1: Check the wiring and regulate it
2: Install the reactor or filter
3: Check whether the fan and the ventilation duct are normal
4: Please contact the supplier for repairing</td><td>3</td></tr>
<tr><td>E0009</td><td>E09</td><td>Heatsink overheated</td><td>1: Ambient temperature exceeds specifications
2: The controller external ventilation is adverse
3: Fan is faulty
4: Temperature detection circuit is faulty</td><td>1: Derated for using and increase power
2: Rectify controller external ventilation
3: Replace the fan
4: Seek for technical support</td><td>3</td></tr>
<tr><td>E0010</td><td>E10</td><td>Braking unit faulty</td><td>The braking circuit is faulty</td><td>Seek for technical support</td><td>3</td></tr>
<tr><td>E0011</td><td>E11</td><td>CPU fault</td><td>CPU is abnormal</td><td>1: Power-on observation after completely power down
2: Seek for technical support</td><td>3</td></tr>
</table>

Panel display	Small panel display	Fault name	Fault cause	Countermeasure	Sort
E0012	E12	Parameter auto-tuning fault	1: Parameter auto-tuning timeout 2: Over current at parameter auto-tuning 3: Under the distance control (set F00.07 as 1) doing permanent magnet synchronous motor rotating auto-tuning (set F10.10 as 2)	1: Check the motor wiring 2: Input correct motor parameters (Group F07 / Group F10) 3: Do the permanent magnet synchronous motor rotating auto-tuning under the panel control (set F00.07 as 0)	3
E0013	E13	Soft start failed	1: Contactor fault 2: Control circuit fault	1: Replace the contactor 2: Seek for technical support	3
E0014	E14	Current detect faulty	1: Current detection circuit damage 2: Permanent magnet synchronous motor is out of control	1: Please contact the supplier for repairing 2: Check the brake signal	3
E0015	E15	Lack of input	For three-phase input controller, three-phase input power phase loss	1: Check the three-phase input power 2: Check the settings of parameter F17.00 and F17.01	3
E0016	E16	Lack of output	1: Controller three-phase output broken or loss of phase 2: Controller with serious imbalance in three-phase load	1: Check the wiring between controller and motor 2: Check the motor 3: Check the settings of parameter F17.02 and F17.03	3
E0017	E17	Controller overloaded	1: Brake circuit abnormal 2: Load is excessive 3: Encoder feedback signal abnormal 4: Motor parameter error 5: Check motor power line	1: Check the brake circuit 2: Reduce the load 3: Check the encoder feedback signal 4: Check the motor parameters and restart the parameter auto-tuning (Group F07 / Group F10) 5: Check the power line	3

Panel display	Small panel display	Fault name	Fault cause	Countermeasure	Sort
E0018	E18	Excessive speed deviation	1: Brake contactor fault or run contactor fault 2: Encoder pulse number setting error 3: Excessive deviation of detection value and time setting unreasonable 4: Controller output torque is not enough 5: Speed-loop PI parameter setting is improper 6: Encoder signal error 7: Motor parameter error 8: F10.12 error	1: Check the brake contactor or the run contactor 2: Reasonably set encoder pulse parameter (F11.01) 3: Correctly set F04.11 (detected value) and F04.12 (detected time) 4: Select larger capacity controller 5: Correctly set speed-loop PI parameter (F08) 6: Check encoder wiring and installation 7: Check the motor parameter 8: Restart parameter auto-tuning	3
E0019	E19	Motor overloaded	1: Brake circuit abnormal 2: Motor overload protect factor set incorrectly 3: Load is excessive	1: Check the brake circuit 2: Correctly set motor overload protect factor (F17.04) 3: Reduce the load	2
E0020	E20	Motor overheated	1: Motor is overheated 2: Motor overheating input signal action 3: Motor parameter setting error	1: Reduce the load 2: Detect whether the overheating input terminal signal is correct 3: Correctly set motor parameter (Group F07 / Group F10)	2
E0021	E21	MCB EEPROM read/write faulty	MCB EEPROM circuit failure	Contact the supplier for repairing	3
E0022	E22	Panel EEPROM read/write faulty	Panel EEPROM circuit failure	1: Replace the panel 2: Contact the supplier for repairing 3: Panel manually reset to continue normal use (exclude parameter upload and download)	1

Panel display	Small panel display	Fault name	Fault cause	Countermeasure	Sort
E0023	E23	Parameter setting faulty	1: In the non-panel mode, asynchronous motor parameter auto-tuning to set parameters auto-tuning 2: Synchronous motor selects ABZ encoder 3: Synchronous motor static auto-tuning, the operating mode is panel setting 4: Motor current is set to zero 5: Asynchronous motor no-load current setting value is larger than motor rated current 6: The creeping speed at distance control (F04.02) is larger than highest speed of running curve (F19.07－F19.11) 7: 0.000m/s＜F19.07－F19.11＜0.100m/s 8: Firefighting base station, locked-elevator base station and idle base station are set to non-service floor 9: Door service floors of firefighting base station, locked-elevator base station and idle base station are set to prohibit service	1: At asynchronous motor parameter auto-tuning, set F00.07 as 0 (panel control) 2: For the synchronous motor, F11.00 (encoder interface card selection) should be set as 2 (UVW encoder interface card) or 3 (SINCOS encoder interface card) 3: At synchronous motor static auto-tuning, set F00.07 as 1 (distance control) 4: Correctly set motor current (F07.02 / F10.03) 5: Correctly set asynchronous motor no-load current (F07.11) 6: Restart to set F04.02, F19.07－F19.11 7: Restart to set F19.07－F19.11 8: Restart to set F21.03, F22.01 and F22.02 9: Restart to setF21.03	3
E0024	E24	Input voltage detection failure	Input normal bus voltage, but the line voltage detection circuit is abnormal	1: Power-down treatment 2: Contact to factory for repairing	1
E0030	E30	Encoder reverse direction	1: The preset speed direction is inconsistent with the actual direction 2: Load is too large 3: Controller output torque is not enough 4: Brake circuit abnormal 5: Run contactor abnormal	1: At elevator commissioning, F11.02 (encoder direction) value is reverse; during normal running, do not modify F11.02 2: Reduce the load 3: Select larger capacity controller 4: Check the brake circuit 5: Check the run contactor	3
E0031	E31	Encoder disconnection	1: Encoder without input signal 2: Brake circuit abnormal	1: Check the encoder wiring and encoder installed reliably 2: Check the brake circuit	3

Panel display	Small panel display	Fault name	Fault cause	Countermeasure	Sort
E0032	E32	Motor over speed	1: Encoder pulse number setting error 2: Controller output torque is not enough 3: Speed-loop PI parameter setting is improper 4: Encoder signal error 5: F10.12 error 6: Motor parameter error	1: Reasonably set encoder P/R (F11.01) 2: Select larger capacity controller 3: Correctly set speed-loop PI parameter (Group F08) 4: Check the encoder wiring and encoder installed reliably 5: Restart parameter auto-tuning 6: Check motor parameter	3
E0033	E33	Loss of Z signal of ABZ encoder	1: Wiring problem 2: Serious interference	Check the wiring	3
E0034	E34	UVW signal wrong of UVW encoder	UVW encoder sector confirmation is wrong	Whether the wiring of UVW is correct	3
E0035	E35	CD phase wrong of Sincos encoder	1: Encoder fault 2: Encoder disconnection	1: Check the encoder 2: Check the wirings of encoder C phase and D phase	3
E0036	E36	Shortest distance ultrahigh	1: Speed curve setting is inappropriate 2: Acceleration/deceleration setting is inappropriate	1: Set appropriate speed curve (F19.07－F19.11) 2: Set appropriate Acc/Dec curve parameters (F03.00－F03.05)	3
E0037	E37	Control board logic parameters	The main control board logic is abnormal	Please contact the supplier for changing the main control board	
E0038	E38	Up forced Dec switch disconnection	Elevator on the top floor, up forced deceleration switch is turned off	1: Check the up forced Dec switch 2: Restart shaft self-learning 3: Check the leveling switch signal	3
E0039	E39	Down forced Dec switch disconnection	Elevator on the first floor, down forced deceleration switch is turned off	1: Check the down forced Dec switch 2: Restart shaft self-learning 3: Check the leveling switch signal	3
E0040	E40	Elevator run timeout	Leveling signal without any change within F23.02 specified time	1: Elevator speed is too low, or floor height is too high 2: Leveling signal is abnormal 3: Steel wire skid	3

Panel display	Small panel display	Fault name	Fault cause	Countermeasure	Sort
E0041	E41	Safety circuit disconnection	Safety circuit signal disconnection	1: Check the safety circuit switch, and view the status 2: Check the safety circuit power supply circuit 3: Check the safety circuit contactor signal 4: Check the safety circuit feedback contact signal characteristics (normally open or normally closed)	3
E0042	E42	Door locked disconnection during running	During elevator running process, the door locked signal is disconnected	1: Check whether the hall and the car door lock contact is normal 2: Check whether the door lock contactor action is normal 3: Check the door lock contactor feedback contact characteristics (normally open or normally closed) 4: Check the door lock power supply circuit 5: If there is MT70-AOB-A, check the corresponding signal	3
E0043	E43	Up limit signal disconnection during running	1: The signal of up limit is cut off when elevator is up running 2: Encoder signal interference makes elevator position error	1: Check that the up limit switch contact is normal or not 2: Check the up limit switch signal characteristics (normally open or normally closed) 3: Up limit switch installed low, normal run to the top will be action 4: Check encoder wiring and installation	3
E0044	E44	Down limit signal disconnection during running	1: The signal of down limit is cut off when elevator is down running 2: Encoder signal interference makes elevator position error	1: Check that the down limit switch contact is normal or not 2: Check the down limit switch signal characteristics (normally open or normally closed) 3: Down limit switch installed high, normal run to the bottom will be action 4: Check encoder wiring and installation	3
E0045	E45	Up/down forced Dec switch disconnection	Up/down forced Dec switches simultaneously disconnected	1: Check whether up/down forced Dec switches are normal 2: up/down forced Dec signal characteristics (normally open or normally closed) 3: F26.12 (inspection parameter setting) of Bit4 is set as 1	3

Panel display	Small panel display	Fault name	Fault cause	Countermeasure	Sort
E0046	E46	Re-leveling abnormal	1: Elevator actual speed is larger than re-leveling speed +0.050m/s 2: Re-levelling position is not in the leveling area	1: Check the encoder signal 2: Check the leveling signal 3: Check the advanced open door block	3
E0047	E47	Lock-door contactor adhesion	Lock-door contactor feedback signal abnormal	1: Check lock-door contactor feedback signal characteristics (normally open or normally closed) 2: Check lock-door contactor action is normal or not 3: Check lock-door contactor feedback signal 4: Check the advanced open door block	3
E0048	E48	OD fault	OD continuous non-arrival times are over F22.09	1: Check the door machine system 2: Check the OD arrival signal is normal or not	3
E0049	E49	CD fault	CD continuous non-arrival times are over F22.09	1: Check the door machine system 2: Check the CD arrival signal is normal or not 3: Check the door lock circuit	3
E0050	E50	Shaft self-learning fault	At the beginning of the learning, if any of the following conditions is met, the fault will be alarmed: 1. The present floor is not the first floor 2. The self-learning direction is not up running 3. Down forced signal is invalid 4. Initial angle of the synchronous motor is 0 5. At two floors, the down leveling sensor isn't out off the leveling plate Run to the second floor, if meet the following condition, alarm fault: At the second floor self-learning, the learned adjustment distance is greater than 50cm	1: Check the up/down forced Dec switch signal 2: Actual floor is consistent with present floor (F19.01) or not 3: Synchronous motor is auto-tuning parameters or not 4: Check whether the motor actual running direction is correct 5: Check whether the leveling plate installation is correct 6: Check whether the leveling switch normally open/closed setting is right	3

Panel display	Small panel display	Fault name	Fault cause	Countermeasure	Sort
E0050	E50	Shaft self-learning fault	Run to the top floor, if meet any of following conditions, alarm fault: 1. Up forced Dec 1 action is valid and in the door zone, and the present floor is inconsistent with the preset maximum floor 2. Elevator reaches the set floor and in the door zone, and the up forced Dec 1 is no action 3. The learned height of total floor is lower than 50cm 4. The learned up/down forced Dec 1 position is 0 5. If configured 2 and 3 level forced Dec switches, the learned up and down forced Dec position is 0 6. If you select multiple forced Dec signal, if does not meet the following conditions, it will alarm fault: Down forced position 1< Down forced position 2< Down forced position 3 Up forced position 1> Up forced position 2> Up forced position 3		3
E0053	E53	Lock-door short-circuit fault	OD arrival signal and lock door closure signal are valid at the same time	1: Check the door lock circuit action is normal or not 2: Check the door lock contactor feedback is normal or not 3: Check the door machine OD arrival signal 4: F26.12 (inspection parameter setting) of Bit3 is set as 1	3
E0054	E54	Synchronous motor star-delta contactor feedback abnormal	Synchronous motor star-delta contactor feedback abnormal	1: Check whether the contactor feedback contact is consistent with MCB parameter setting (normally open or normally closed) 2: Check whether the indicator on MCB output side is consistent with contactor action	3

Panel display	Small panel display	Fault name	Fault cause	Countermeasure	Sort
E0054	E54	Synchronous motor star-delta contactor feedback abnormal	Synchronous motor star-delta contactor feedback abnormal	3: After the contactor acts, check whether the corresponding feedback contact and MCB corresponding feedback input point acts 4: Check whether the output characteristics of contactor is consistent with that of MCB 5: Check the contactor coil circuit	3
E0055	E55	Changed floor park fault	When elevator runs automatically, the floor has not received OD arrival signal	1: Check the door machine OD arrival signal 2: Check the door mechanical system	1
E0056	E56	Run contactor feedback abnormal	Run contactor feedback abnormal	1: Check whether the contactor feedback contact is consistent with MCB parameter setting (normally open or normally closed) 2: Check whether the indicator on MCB output side is consistent with contactor action 3: After the contactor acts, check whether the corresponding feedback contact and MCB corresponding feedback input point acts 4: Check whether the output characteristics of contactor is consistent with that of MCB 5: Check the contactor coil circuit 6: F26.17 is set as 1, the fault will restore	3
E0057	E57	Brake contactor feedback abnormal	1: Brake contactor feedback signal abnormal 2: Brake mechanical switch feedback abnormal 3: Brake forced feedback abnormal	1: Check whether the contactor feedback contact is consistent with MCB parameter setting (normally open or normally closed) 2: Check whether the indicator on MCB output side is consistent with contactor action 3: After the contactor acts, check whether the corresponding feedback contact and MCB corresponding feedback input point acts 4: Check whether the output characteristics of contactor is consistent with that of MCB	3

Panel display	Small panel display	Fault name	Fault cause	Countermeasure	Sort
E0057	E57	Brake contactor feedback abnormal	1: Brake contactor feedback signal abnormal 2: Brake mechanical switch feedback abnormal 3: Brake forced feedback abnormal	5: Check the contactor coil circuit 6: Check the brake mechanical switch feedback signal 7: Check the brake forced feedback signal 8: Check the brake forced contactor coil 9: F26.17 is set as 1, the fault will restore	3
E0058	E58	Leveling signal abnormal	Leveling/door zone signal is adhesion or cut off	1: Check whether the leveling and the door zone can work normally 2: Check the vertical and depth of leveling plate installation 3: Check the MCB input point	3
E0059	E59	Receive OD and CD arrival signals at the same time	Receive door machine OD and CD arrival signals at the same time	1: Check the door machine controller 2: Check OD/CD arrival signal characteristics (normally open or normally closed) 3: At inspection mode, F26.12 (inspection parameter setting) of Bit5 is set as 1, which can shield the fault	3
E0060	E60	Forced Dec distance is too short	Forced Dec distance is too short	1: Check up/down forced Dec 1 switch installation 2: Check the forced Dec speed (F03.12)	3
E0062	E62	Inspection run overcurrent	Inspection running current is 110% over motor rated current	1: Reduce the load 2: F26.12 of Bit1 is set as zero 3: Permanent magnet synchronous motor identified the encoder angle does not match with the actual, restart parameter auto-tuning 4: Encoder abnormal 5: Brake circuit abnormal	3
E0063	E63	Advanced open door abnormal	1: Speed is larger than advanced open speed + 0.050m/s 2: Advanced open operation is not in the leveling	1: Check the encoder signal 2: Check the leveling signal 3: Check the advanced open block (MT70-AOB-A)	3

The following faults can automatically reset:

1. E0009 heatsink overheated fault: After the heatsink temperature drops to 50℃, the fault will reset automatically.

2. E0020 motor overheated fault: After motor overheated switch recovers, the fault will reset automatically.

3. E0041 safety circuit disconnection fault: After the safety circuit is connected, the fault will reset automatically.

4. E0042 door locked disconnection fault: After locked-door is connected and auto reset, or door zone signal is valid, 1s later, the fault will reset automatically.

5. E0055 changed floor park fault: The fault only recorded once at power-on.

6. E0059 OD and CD arrival signals at the same time fault: The fault only recorded once at power-on, and if OD/CD arrival signals are not valid at the same time, the fault will reset automatically.

7. E0048, E0049 and E0055 faults can be reset by inspection button.